

Experiment 8

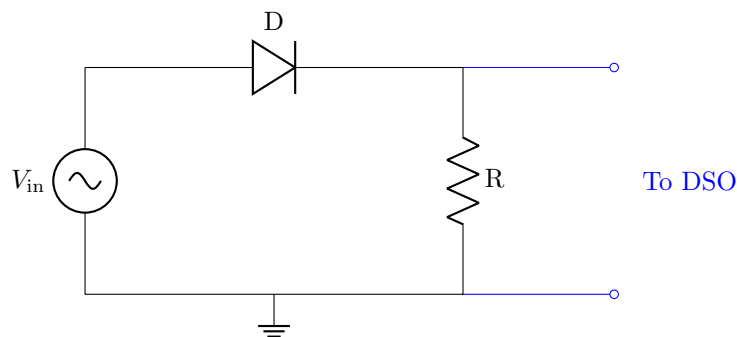
Diode circuits

Components/Instruments/Meters

- Digital storage oscilloscope (DSO)
- Function generator
- DC power supply
- Breadboard
- Resistors
- Capacitors
- Diodes
- Connecting wires

Part 1: Half wave rectifier

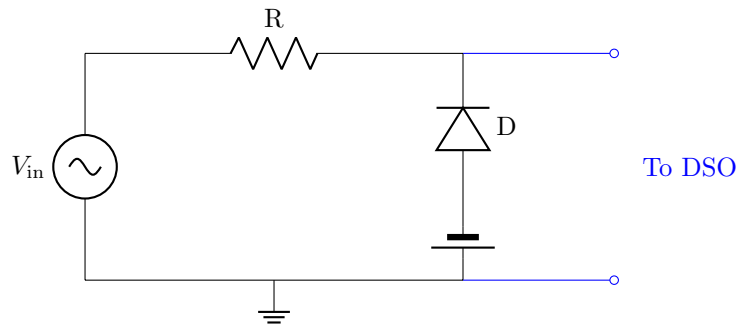
1. Setup a circuit as shown below. Take the value of R as 1 k Ω .



2. Set the signal generator to sine wave signal (Peak-to-peak value = 2 V, frequency = 100 Hz).
3. Connect 'Channel-1' of DSO to input terminals of the circuit.
4. Connect 'Channel-2' of DSO to observe the voltage across the resistor.
5. Vary the amplitude of the input signal in steps of 1 V till 5 V, and measure the amplitude of the output voltages.

Part 2: Clipper

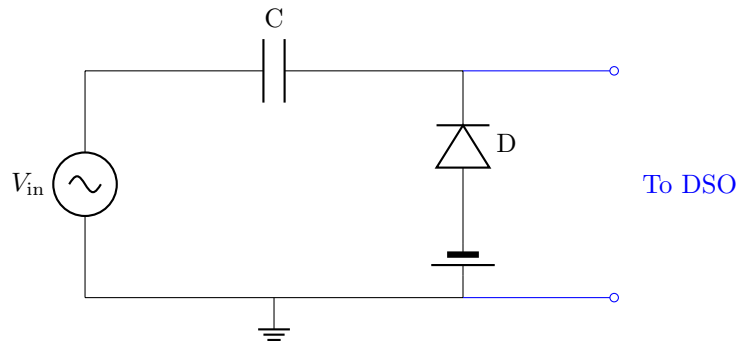
1. Setup a circuit as shown below. Take the value of R as $1\text{ k}\Omega$.



2. Set the signal generator to sine wave signal (Peak-to-peak value = 5 V , frequency = 100 Hz).
3. Connect 'Channel-1' of DSO to input terminals of the circuit.
4. Connect 'Channel-2' of DSO to observe the output voltage.
5. Vary the DC voltage from 0 V to 1 V and then 2 V .
6. Observe the input and output waveforms and record these waveforms.
7. Are you able to explain the output waveforms?

Part 3: Clamper

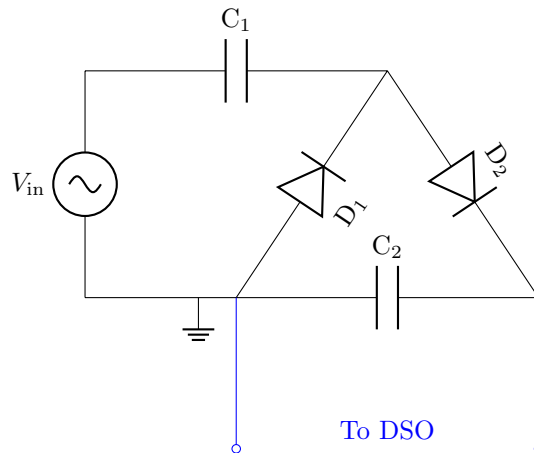
1. Setup a circuit as shown below. Take the value of C as $1\ \mu\text{F}$.



2. Set the signal generator to sine wave signal (Peak-to-peak value = 5 V, frequency = 100 Hz).
3. Connect 'Channel-1' of DSO to input terminals of the circuit.
4. Connect 'Channel-2' of DSO to observe the output voltage.
5. Vary the DC voltage from 0 V to 1 V and then 2 V.
6. Observe the input and output waveforms and record these waveforms.
7. Are you able to explain the output waveforms?

Part 4: Voltage multiplier

1. Setup a circuit as shown below. Take all the capacitors to be $1\ \mu\text{F}$.



2. Set the signal generator to sine wave signal (Peak-to-peak value = 5 V, frequency = 100 Hz).
3. Connect 'Channel-1' of DSO to input terminals of the circuit.
4. Connect 'Channel-2' of DSO to observe the output voltage.
5. Observe the input and output waveforms and record these waveforms.

Optional: Based on your observations, could you design a circuit where the output voltage is approximately 4 times that of the peak input voltage? If yes, try to design and validate it.